MILITARY SPECIFICATION

SEMITRAILER, TANK: AIRCRAFT REFUELING, 5,000 GALLON, TURBINE FUEL AND AVIATION GASOLINE, COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of 5,000 gallon, tandem axle, aircraft fuel servicing tank semitrailers with a 300 or 600 gallon per minute (gpm) fuel servicing rate. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in the procurement documents.

* 1.2 Classification. The semitrailer shall be one of the following types, as specified (see 6.2):

Type I - 300 gpm refueling rate
Type II - 600 gpm refueling rate.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GBS.1, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-2330

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FEDERAL

O-E-915	- Extinguisher, Fire, Dry-Chemical, (Hand-Portable).

MILITARY

MIL-G-5572	- Gasoline, Aviation, Grades 80/87, 100/130, 115/145.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-N-5877	- Nozzle, Pressure Fuel Servicing, Locking, Type D-1, Nominal 2 1/2 Inch Diameter.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-F-8901	- Filter-Separators, Liquid Fuel: and Filter-Coalescer Elements, Fluid Pressure: Inspection Requirements and Test Procedures For.
MIL-C-10387	- Couplng, Clamp Pipe, with Bolts and Synthetic Rubber Gasket, for Grooved Pipe and Tube.
MIL-P-10388	- Pipe Fitting, One or More Ends Grooved.
MIL-A-27302	- Arrester, Spark, Exhaust, Internal Combustion Engine.
MIL-C-27487	- Coupling Halves, Quick-Disconnect, Cam-Locking Type.
MIL-F-52308	- Filter Element, Fluid Pressure.
MIL-N-52747	- Nozzle Assembly, Closed-Circuit Refueling.
MIL-M-81380	- Monitor, Contamination, Fuel Dispensing System.
MIL-C-83413	- Connector, Electricl Ground.
MIL-N-87963	- Nozzle: Fuel, Aircraft, Overwing: Type MD-3.

STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-1223	- Non-Tactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification Marking and Data Plate Standards.
MS75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24-Volt, Waterproof.

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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.
Regulations Governing Transportation of Dangerous Articles in Tank Motor Vehicles (49 CFR 178).

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

NATIONAL BUREAU OF STANDARDS
Handbook 44 - Specifications and Regulations for Commercial Weighing and Measuring Devices.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN PETROLEUM INSTITUTE (API)
RP 1004 - Bottom Loading and Vapor Recovery for MC306 Tank Motor Vehicles.
RP 2003 - Recommended Practice for Protection Against Ignitions Arising out of Static, Lightning and Stray Currents.

(Application for copies should be addressed to the American Petroleum Institute, 2101 L Street, N.W., Washington, D.C. 20037.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Boilers and Pressure Vessel Code - Section VIII.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
NFPA No. 407 - Aircraft Fuel Servicing 1980.

(Application for copies should be addressed to National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

SAE, INC.
SAE Standards and Recommended Practices.

J318	- Airbrake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers.
J551	- Limits and Methods of Measurement of Radio Interference Characteristics of Vehicles and Devices (20-1000 MHz) (DoD adopted).
J552	- External Electromagnetic Radiation Suppressors.
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J682	- Rear Wheel Splash and Stone Throw Protection.
J700	- Fifth Wheel Kingpin - Commercial Trailers and Semitrailers.
J702	- Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer.

(Application for copies should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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STATE OF CALIFORNIA
Air Resources Board
Executive Order G-70-10-A

(Application for copies should be addressed to the Air Resources Board, 1102 Q Street, P.O. Box 2815, Sacramento, CA 95812.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. Except as otherwise specified herein, in the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard semitrailer, components, and accessories. Except as specified in 3.1.1 through 3.1.1.4, the semitrailer, components, and accessories shall be standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented and rated in the manufacturer's technical data, including special or mounted equipment. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of semitrailer models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items. The semitrailer and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.1.1 Special requirements. In addition to the standard semitrailer and components specified in 3.1, the semitrailer shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, identification marking and data plates shall be in accordance with MIL-STD-1223 except the exterior color shall be gloss yellow, matching color chip No. 13538 of FED. STD. No. 595. In addition, the vehicle body shall be marked in accordance with paragraphs 4-11 through 4-11.2 of NFPA No. 407.

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* 3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with MIL-STD-1223. When specified, (see 6.2), tropical rustproofing in accordance with MIL-STD-1223 shall be furnished.

3.1.1.3 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Suppressors used shall conform to SAE J552.

3.1.1.4 Drain plugs. All drain plugs installed throughout the vehicle for the purpose of draining lubricants shall be of the permanent magnet type.

3.2 Weights, loads and dimensions.

3.2.1 Net weight. The net weight of the fully equipped semitrailer shall be the minimum practicable for the service intended.

* 3.2.2 Rated payload capacity. The rated payload capacity shall be not less than the weight computed from the rated liquid capacity (5,000 gallons), using 6.8 pounds per gallon for the product (turbine fuel).

3.2.3 Gross weight. The gross weight shall consist of the net weight and the payload.

* 3.2.4 Weight distribution. Load bearing members of the semitrailer shall be rated for the loads imposed with the tank filled to nominal capacity, and a maximum tank slope of 1 percent to the rear.

* 3.2.5 Dimensions and clearances. The semitrailer, uncoupled from the towing vehicle, resting level on the landing legs, without payload, and on level ground, shall conform to the dimensions specified in table I.

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TABLE I. Dimensions and clearances.

Characteristic	Inches
Overall length (maximum)	384
Overall width (maximum)	96
Overall height (maximum)	105
Minimum swing clearance (radius from center of the kingpin to any point 6 inches below upper fifth wheel plate) (not less than)	82
Swing radius from the centerline of the kingpin to the most distant point on the semitrailer nose (not more than)	56
Distance from the centerline of kingpin to front end of the semitrailer (not more than)	36
Upper fifth wheel height (plus or minus one inch)	48
Ground clearance (not less than)	10
Landing leg ground clearance (not less than)	11

3.3 Performance requirements.

* 3.3.1 Functional performance. The semitrailer shall be capable of transporting 5,000 gallons of aircraft turbine fuel and shall be capable of performing the fuel servicing functions specified in 3.6.14.3. The semitrailer shall be capable of all fuel transfer functions while either connected to or separated from the towing vehicle.

* 3.3.2 Mobility performance. The semitrailer, fully equipped and loaded with the rated payload, shall be capable of being towed at speeds of not less than 10 miles per hour (mph) over unimproved roads (see 6.3), and over reasonably hard uneven terrain, and of being towed at speeds of not less than 55 mph over improved roads.

3.3.2.1 Turning ability. The semitrailer shall be capable of passing through a 90 degree angle to the towing vehicle without cramping or damage to the semitrailer or the towing vehicle.

3.3.2 Tracking ability. The semitrailer shall conform to the tracking ability requirements of Federal Motor Carrier Safety Regulation 393.70(a).

* 3.3.2.3 Brake performance. The service brakes shall stop the semitrailer-tractor combination, with the semitrailer loaded with specified payload, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

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3.4 Chassis components.

3.4.1 Chassis. The chassis frame shall consist of the tank shell (with its internal bulkheads) as the primary frame member, with welded external longitudinal members connected by crossmembers distributing concentrated loads into the shell. All structural design shall be in accordance with 49 CFR, sections 178.340 and 178.341. The primary tank structure shall support the payload and provide attachment for the upper fifth wheel plate and kingpin, suspension system, landing gear, underslung cabinets, and associated piping and accessories.

3.4.2 Suspension system. The semitrailer shall be furnished with the manufacturer's standard suspension system. Each component of the suspension system shall have a rated capacity at least equal to the load imposed, measured at the wheels at the ground, when the semitrailer is loaded with its rated payload (see 3.2.2). Clearance shall preclude interference between wheels and with any other part of the semitrailer under the operating conditions specified herein.

3.4.3 Tandem axles. Tandem axles shall be furnished. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, when the semitrailer is loaded with the rated payload. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hub caps shall have a window for visual determination of the oil level. Provision for venting, or other method of withstanding internal pressure buildup, and for replenishing the oil supply, shall be incorporated. The hubs shall be fitted with seals.

* 3.4.4 Wheels, rims, tires and tubes. The semitrailer shall be equipped with dual wheels on each axle. Rim and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. Rim sizes shall be the same for all wheels on the semitrailer. Tire size and load range (ply rating) shall be the same for all tires on the semitrailer. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.4.1 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be of the steel belted radial type or, when specified (see 6.2), bias ply. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the semitrailer loaded with specified payload.

* 3.4.4.2 Inner tubes. When tube-type tires are furnished, the inner tubes shall be of the heavy duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided in accordance with Tire and Rim Association recommendations.

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* 3.4.4.3 Carrier for spare tire assembly. Unless otherwise specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location. The carrier shall permit removal or installation of a spare wheel assembly, using only tools furnished with the semitrailer.

* 3.4.4.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be mounted and secured on the carrier.

* 3.4.4.5 Spare tire. When specified (see 6.2), an inflated spare tire shall be mounted on the spare wheel or rim. The spare tire shall be of the same size, load range, and tread design as the tires installed on the semitrailer.

3.4.5 Brakes.

* 3.4.5.1 Service brakes. The service brakes shall be of the full air, internal expanding type, and shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43, 393.45 through 393.47, and 393.52. The braking system shall include standard breakaway features, relay emergency valve, air reservoirs, slack adjusters, piping, hose connections, gladhands, spring loaded dust covers or dummy gladhands equipped with security chains, and all other components required for a complete air brake system. The location of the gladhands shall comply with SAE J702. The gladhands shall conform to SAE J318. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by objects striking the components. No part of the braking system shall extend below the bottom of the wheel rims. The air reservoirs shall be of sufficient capacity to supply air for the air brakes and the complete tank pneumatic control system of the semitrailer, with an adequate reserve capacity.

* 3.4.5.2 Parking brakes. The semitrailer shall be equipped with spring applied, or air diaphragm type, parking brakes which shall hold the semitrailer, with rated payload, on a 10 percent grade despite the depletion of the compressed air supply. The parking brake shall be automatically applied upon disconnection of the emergency air line, and under emergency braking conditions, and shall remain applied without any further application. The parking brake shall also be automatically applied (interlocked with the nozzles and bottom loading adapter) during all fuel handling operations as specified herein.

3.4.6 Upper fifth wheel plate. The upper fifth wheel plate shall be designed for coupling to a full oscillating and fore-and-aft rocking

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fifth wheel, shall be of sufficient size to cover a fifth wheel 36 inches in diameter, and shall conform to Federal Motor Carrier Safety Regulation 393.70(b). The kingpin shall be fabricated of heat treated alloy steel and shall conform to SAE J700. The forward end of the upper fifth wheel plate or skid plate shall have a turned-up lip for ease of loading and for body protection.

3.4.7 Electrical equipment. All electrical equipment shall conform to paragraph 4-3.8 of NFPA No. 407.

3.4.7.1 Twelve volt dc lighting system. A 12-volt direct current (dc) lighting system shall be provided. The 12-volt dc lighting system shall conform to Federal Motor Carrier Safety Regulations 393.14, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of the rub rails or the semitrailer bumper. Turn signal lamps shall conform to SAE J588.

3.4.7.2 Receptacle, 12-volt dc. The front of the semitrailer shall be equipped with a 7-contact receptacle conforming to SAE J560 with the connectors connected and color coded as specified therein. The receptacle shall be provided with a spring loaded cover. The receptacle shall be located in accordance with SAE J702.

3.4.7.3 Independent 24-volt dc lighting system. When specified (see 6.2), a separate 24-volt dc lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.4.7. The 24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.4.7.1, except the 24-volt dc stop and turn lamps shall be combined.

* 3.4.7.4 Receptacle, 24-volt dc. When an independent 24-volt dc system is specified (see 3.4.7.3), the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B.	Connect to left-hand turn signal and stop lamp (yellow).
Contact D.	Connect to ground (white).
Contact E.	Connect to clearance, side marker, identification, and tail lamps (black and brown).

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Contact J. Connect to right-hand turn signal and stop lamp (green).
Contact L. Connect to ground (white).

The rest of the contacts shall not be connected. Circuits B and J on tactical trucks are combination stop and turn indicator circuits.

* 3.4.7.5 Circuit protection. Explosion-proof overload circuit breakers shall be used in lieu of replaceable, fuse-type devices on all circuits. Circuit protection devices shall be connected and identified in an organized manner to facilitate determination of malfunction.

* 3.4.7.6 Night servicing lights. Unless otherwise specified (see 6.2), the semitrailer shall be provided with lighting for night servicing of aircraft. The night servicing lights shall include floodlights, cabinet lights, and panel lights and shall be powered by the 12-volt system of the pump engine. The floodlights shall provide illumination of any area accessible to the fuel servicing hoses and shall not extend above the silhouette of the semitrailer. The switches for all night servicing lights shall be located on the control panel.

* 3.4.8 Landing gear. The semitrailer shall have two vertical-lift, telescopic, nonrotating landing legs, with two-speed gears and a handcrank on the curb side. The landing legs shall be equipped with steel wheels or self-leveling skid pads. When specified (see 6.2), both legs shall be capable of being simultaneously operated from either side of the semitrailer. Supports for the crank extension shafts and clips for holding the cranks when folded shall be provided. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to the proportion of gross weight sustained and the forces resulting from impact during coupling and uncoupling operations. The landing gear assembly shall be mounted to plates that are gusseted to resist forward and rearward dynamic thrust on the landing legs. When placed in the travel position, the landing gear legs shall remain positively locked. The landing gear shall be protected to preclude the entrance of foreign matter which would impair its functional or mechanical efficiency. The landing gear shall have a range of adjustment to vary the height of the upper fifth wheel from 47 inches to not less than 52 inches from the ground. With the semitrailer coupled to a towing tractor and in a level position, the clearance under the fully retracted landing gear shall exceed the semitrailer ground clearance, but shall in no case be less than 11 inches.

* 3.4.9 Rear end protection. Semitrailer rear end protection shall be in accordance with Federal Motor Carrier Safety Regulation 393.86, and shall conform to the requirements of 49 CFR, section 178.340-8(b).

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* 3.4.10 Rear wheel splash and stone throw protection. The rear wheels shall have mud flaps at the rear. Splash and stone throw protection shall be in accordance with SAE J682.

* 3.4.11 Tools. Such tools as are normally required for operation of the equipment shall be furnished and stowed. When a spare tire is specified (see 3.4.4.5), tire changing tools shall be furnished and shall include a wheel-nut wrench and hydraulic jack. Hold-down spring type clamps for the tools shall be furnished.

3.5 Stress analysis. The contractor shall provide a stress analysis. This analysis shall verify compliance of the chassis and tank structure with 49 CFR, sections 178.340-4 and 178.341-2 of MC306, and shall include at least the following calculations and diagrams:

(a) The sprung gross weight load distribution diagram.
(b) The vertical shear diagram for the length of the vehicle.
(c) The bending moment diagram for the length of the vehicle.
(d) The loads, sections, and stresses in members that carry maximum shear and moment, to include the analysis of the stresses in attaching weldments or other fastenings.
(e) The loads, sections, shear, and bending moment diagrams at the kingpin and rear underconstruction subframe members, and for the critical section of the tank shell structure.
(f) The load, section, shear, and bending moment diagram for the most critical crossmember, to include the analysis of stresses in attaching weldments or other fastenings.
(g) The loads, sections, and stresses in members that carry maximum compression, and the factor of safety against buckling for those members.
(h) A listing of all assumptions and references utilized to provide required data.

* 3.6 Tank. The tank shall be a cargo tank for transporting aviation turbine fuel conforming to MIL-T-5624 and aviation gasoline conforming to MIL-G-5572. The tank shall be of the single compartment type, with a capacity of not less than 5,000 gallons plus at least 3 percent expansion space. The tank shall have a low profile shape with a modified elliptical cross section. The tank shall conform to 49 CFR, sections 178.340 and 178.341 of MC 306. The tank shall be fabricated of type 304 stainless steel in accordance with 49 CFR, section 178.340-3, except that mild steel, high strength low alloy steel, or combinations of these steels with austenitic stainless steel shall not be furnished. All tank filler openings and vents shall be protected against overturn damage by a rigid member or members firmly fixed to the tank in accordance with 49 CFR, section 178.340-8. Where dissimilar metals are

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used, effective means shall be taken to impede electrolytic action. Teflon packing and tape shall not be used on adjacent parts that are of aluminum content. The tank and all associated fittings, components and piping shall be free of all dirt, loose scale and petroleum soluble coating.

3.6.1 Electrostatic discharge protection. Electrostatic discharge protection, conforming to paragraph 4-3.2 of NFPA No. 407, shall be provided.

* 3.6.1.1 Bonding straps. Bonding straps or bolted metal-to-metal conductive connections shall be provided between the pump, meter, filter/separator or other static producing devices and the nearest structural part of the body or chassis. Bonding straps or bolted connections shall be attached using corrosion resistant bolts and nuts, and using two internal-external locking washers at each attaching point to assure positive electrical connection.

* 3.6.1.2 Electrostatic charge relaxation. Provision shall be made to dissipate electrostatic electrical charges generated in the product fuel during filtration. An adequate charge relaxation time (fuel-to-metal contact) shall be provided before discharge of fuel, equal to the recommendations of API RP 2003.

* 3.6.2 Hydrodynamic loading. The tank shall be capable of being filled at a rate of 600 gallons per minute (gpm) through a 4-inch pipe in the tank manhole, without damage, and at the rate of 600 gpm when bottom loaded through the bottom loading adapter and automatic flow control valve.

* 3.6.3 Static air pressure. The tank, without fuel, shall withstand an internal static air pressure of at least 5 pounds per square inch gage (psig) without evidence of leaking, in conformance with paragraph 4-10.1 of NFPA No. 407.

3.6.4 Sump. A sump shall be provided at the lowest point of the tank to allow accumulation of solid contaminants and water. The tank shall be so designed that water will drain into the sump and not be trapped by the baffles or stand in pockets. A sump drain valve, of not less than 1-inch size, with remote control and extended drain line shall be provided. The sump drain valve shall be protected against flying objects and shall be easily accessible for maintenance.

3.6.5 Baffles. Where the provisions of 49 CFR, section 178.340 would require circumferential reinforcements, baffles shall be used in lieu of ring stiffeners. The baffles shall be flanged and dished. A

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half round or full round opening, of not less than 3-inch radius, shall be provided at the top and bottom of the baffle. Each baffle shall also be provided with not less than an 18-inch diameter manway opening. The edges of the half round or full round openings and the manway opening shall have a flange of approximately 1 inch. The cross sectional area of each baffle shall be not less than 80 percent of the cross sectional area of the tank.

3.6.6 Tank support. The tank shall be properly structurally supported in accordance with the loading considerations and stress levels of 49 CFR, section 178.340-6(b).

* 3.6.7 Manhole. The tank shall have a manhole, located at the middle of the top of tank, with a minimum diameter of 20 inches. A lightweight, liquidtight, easy opening manhole cover, or bolted cover with a liquidtight easy opening fill cover, of not less than 10-inch diameter, shall be provided. The manhole shall be fully compatible with the use of a Parker-Hannifin Model F428A vapor recovery adapter cover (to be added by the government only when top loading). If the cover is hinged, the hinge shall be forward-mounted and the cover shall have self-latching catches. All materials used shall be resistant to corrosion and to deterioriation from contact with the product fuel. Normal breathing venting may be incorporated into the manhole cover (see 3.6.8). The manhole cover or fill cover shall be bonded to the tank body to eliminate the possibility of a spark occurring during opening or closing of the cover. The manhole shall be equipped with a buna-N gasket, or equivalent, to insure a fuel tight seal between the collar and the cover. The entire manhole cover shall be removable for entrance into the tank by removing a bolt-on clamping ring. Safety devices to prevent the manhole or fill cover from opening fully when internal pressure is present shall be provided. When specified (see 6.2), a 2-inch National Pipe Thread (NPT) coupling and plug shall be provided in the tank, near the manhole cover, to permit future installation of an additional fuel level sensing device.

* 3.6.7.1 Tank calibration marker. A fixed marker, visible through the manhole, shall be provided inside the tank to indicate whether the tank is filled to rated capacity. The marker shall be located so as to not interfere with the vapor recovery adapter cover specified in 3.6.7.

* 3.6.8 Vents. Tank vents shall be provided to prevent excessive pressure from building up under both normal and emergency conditions. The vents shall be designed in accordance with 49 CFR, section 178.341-4 of MC 306.

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* 3.6.8.1 Normal venting flow capacity. The normal vents shall have sufficient vapor flow capacity to permit bottom loading at a rate of 600 gpm, and to permit loading and unloading through the plumbing system at 300 gpm for type I semitrailers and 600 gpm for type II semitrailers. Loading and unloading shall be accomplished with the manhole and fill cover closed, without damage and without exceeding the vacuum and pressure limitations of 49 CFR, section 178.341-4 of MC 306 (maximum pressure of 3 psig; maximum vacuum of 1 psig).

* 3.6.8.2 Emergency venting flow capacity. The vapor flow capacity of the emergency venting system, whether through pressure-actuated or fusible venting, shall be not less than that required by 49 CFR, section 178.341-4 of MC 306 for compartment pressures of not more than 5 psig differential. There shall be no steady stream leakage of liquid fuel under full tank head pressure should the vehicle upset with the vent position from vertical or up to 100 degrees in either direciton.

* 3.6.8.3 Air-operated vent(s). Vent(s) shall be equipped for air-piston operation. The air-operated vent(s) shall be spring-loaded shut to prevent the spillage of fuel and vapors. The vent(s) shall be mounted in the top of the tank and as near the center as possible. The air-piston(s) shall be capable of exerting a force of at least 150 pounds in order to break the vent loose under icing conditions.

3.6.8.4 Vent construction. The vent valves shall be of aluminum with a corrosion resistant steel weld ring with the aluminum isolated from contact with steel, or shall be corrosion resistant steel. The vent valves shall be furnished with an interlock to assure that the vent is open during loading and off-loading operations. Tetrafluoroethylene coated valve seats shall be provided for cold weather operation.

* 3.6.8.5 Normal venting pressure limits. When specified for vehicles dispensing aviation gasoline (see 6.2), the lower opening limits for normal venting (inbreathing, outbreathing) shall comply with the opening limits required by the California Air Resources Board in Executive Order G-70-10-A for vapor recovery, as summarized herein:

(a) The entire tank and venting system shall be sufficiently vaportight such that a pressure change of not more than 1 inch of water shall occur within 5 minutes when the tank is pressurized to 18 inches of water (gage) or evacuated to 6 inches of water (gage);
(b) The outbreathing pressure vents shall remain closed as tank compartment pressure is increased above 18 inches of water (gage), but shall open before compartment pressure reaches 27.7 inches of water (1 psig);

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(c) The inbreathing vacuum vents shall remain closed until tank compartment vacuum drops below 6 inches of water, but shall open before a vacuum of 10.4 inches of water is reached (6 ounces per square inch gage).

3.6.9 Catwalk and access ladder. A catwalk shall be provided over the full length of the top of the tank and shall be coated with an abrasive type of paint (ship's grip). Liquidtight flashing shall be installed around the catwalk. The flashing shall be provided with four drain holes, one on each side of the flashing at the front and rear of the catwalk. An access ladder shall be provided.

3.6.10 Underslung cabinets. Underslung cabinets shall be provided on both sides of the tank and shall be of sufficient size and strength to house the engine and pump, filter/separator and contamination monitoring device, the plumbing system, meter, controls, and hose reels. Provisions for stowing and protecting delivery nozzles shall be furnished in the cabinets above the hose reels. The cabinets shall provide for ventilation and drainage in accordance with paragraph 4-3.9 of NFPA No. 407.

* 3.6.10.1 Hose tubes. Enclosed hose tubes shall be provided for four 3-inch by 10-foot hoses. The hose tubes may be incorporated into the tank subframe structure. The hose tubes shall be drainable and provided with hinged covers.

* 3.6.11 Bottom loading system. The tank shall be capable of being bottom loaded at a rate of not less than 600 gpm. To provide bottom loading capability, the tank shall be furnished with an automatic vent valve, a level sensor controlled automatic flow control valve, and a bottom loading adapter with check valve. The system, including the automatic vent, shall be designed fail-safe to prevent damage to the semitrailer or any of its components if the bottom loading system malfunctions or is operated incorrectly. The system shall comply with paragraph 4-14.3 of NFPA No. 407.

* 3.6.11.1 Inlet manifold. The bottom loading inlet manifold shall extend from the bottom loading adapter to the automatic flow control valve in the bottom of the tank. The complete inlet manifold system shall be sized to meet the required system filling rates and pressures.

* 3.6.11.2 Automatic flow control valve. An automatic flow control valve which serves as a combination bottom loading and emergency valve shall control the flow of fuel into the tank during bottom loading. The control valve shall be automatically controlled and shut off flow when the tank has reached its rated capacity. A remotely operated manual

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control shall be provided also to open or close the valve. This remote control valve operation shall be in simultaneous actuation with the air-operated vent (see 3.6.8.3). The design of the control valve shall be in accordance with the emergency flow control type valve as described in 49 CFR, section 178.341-5 of MC 306, for assurance against accidental escape of contents. Automatic shut off of the flow control valve during bottom loading operation shall be actuated by a tank level sensor that shall signal when the tank has reached its rated fluid capacity (see 3.6.11.2.1). A deflector, in accordance with paragraph 4-14.3.8 of NFPA No. 407, shall be mounted in the tank over the control valve in order to prevent the fuel from spraying unrestricted into the tank.

3.6.11.2.1 Tank level-sensor control. The level control shall automatically shut off the automatic flow control valve when the tank is filled with 5,000 gallons of fuel. The level control unit may be a hydraulic jet type sensing device utilizing line fuel pressure with a jet stream directed to a receiver. A means shall be provided to precheck the level control system to determine if it is functioning properly. The level sensing device shall be fail-safe in case of line breakage or stoppage. The level control system shall operate properly within the designed system pressure range and shall operate with the type of fuel specified without causing undue corrosion or deterioration of the equipment. The level sensor and any associated valves shall be free of all liquids with a freezing point above minus 50°F.

* 3.6.11.3 Bottom loading adapter. Unless otherwise specified (see 6.2), the bottom loading adapter shall be of the "Y" type capable of connection to both a four-inch coupler conforming to American Petroleum Institute (API) RP 1004 and to a 2 1/2-inch type D-1 nozzle conforming to MIL-N-5877. The adapter shall have a dry break feature and shall be on the right (curb) side of the vehicle. The adapter shall be furnished with a bonded nose seal. The adapter shall be provided with a poppet type, spring loaded check valve. The valve shall be full swiveling and free floating, forming a fuel resistant seal to prevent leakage. Dust caps, complete with security cables or chains, shall be provided for the adapters. All parts of the adapters, valves, caps and chains shall be corrosion resistant. Non-electrical interference type interlocks shall be provided to prevent movement of the vehicle when either leg of the bottom loading adapter is engaged for loading.

3.6.11.4 Pumping system interlock. An interlock shall be provided in the pumping system that will prevent fuel from being pumped into or out of the tank unless the air-operated vent (see 3.6.8.3) is open. The interlock shall also automatically shut down the pumping system in the event the vent closes. The interlock shall function without allowing the internal tank pressure to exceed minus 1 or plus 3 psig. The

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auxiliary vents shall not be considered in meeting this requirement. The interlock shall be incorporated into the pumping system controls.

* 3.6.12 Self-loading capability. The semitrailer shall be capable of self-loading (loading by use of on-board pumping system) at a rate of not less than 300 gpm (see 3.6.14.3.1(g)).

* 3.6.13 Vapor recovery system. The air-operated vents (see 3.6.8.3) shall be furnished with a vapor recovery hood or collecting chamber and manifolded to a rollover rail or other suitable manifold piping. Connections between the tank vapor space vents and the header tube(s) shall be not less than 7 square inches in open cross sectional area. The header tube shall be not less than 3 inches in diameter (7 square inches in open cross sectional area). The manifold piping shall terminate at the rear of the semitrailer and unless otherwise specified, shall be provided with an American Petroleum Institute standard 4-inch cam and groove vapor recovery connector conforming to API RP 1004. When specified (see 6.2), the manifold piping shall terminate with a 3-inch cam and groove coupling conforming to MIL-C-27487. The connector shall be provided with a screen or open mesh type camlock cap with security chain. The connector, cap and chain shall be corrosion resistant.

* 3.6.13.1 Vapor recovery connector locations. The vapor recovery connector shall be located on the curbside of the vehicle between 2.5 and 5 feet above ground level.

* 3.6.13.2 Vapor discharge outlet. Provision shall be made for discharging to atmosphere vapors from the vapor recovery system when vapor collection facilities are not available. The location for the outlet shall be on the top of the vehicle, remote from the driver's loading area, and isolated from the truck-tractor engine and pump engine exhaust outlets.

* 3.6.13.3 Vapor recovery system protection. Any portion of the vapor recovery system installed on the top of the tank which could damage the tank in the event of overturn shall be protected by a rollover device. Any portion of the vapor recovery system installed at the rear of the tank shall not extend beyond the rear bumper.

3.6.14 Engine driven pump system. An engine driven pumping system shall be provided. The engine and pump shall be readily accessible for servicing.

* 3.6.14.1 Engine. Unless otherwise specified (see 6.2), a liquid-cooled four stroke cycle gasoline engine shall be provided to drive the pump, an air compressor, and a 12-volt alternator. When

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specified (see 6.2), a diesel engine shall be provided. The engine shall have adequate horsepower, torque and speed characteristics to meet satisfactorily all pumping performance requirements specified in 3.6.14.3, the air compressor requirements specified in 3.6.14.8, and full alternator load. During simultaneous maximum pump rate, maximum compressor load, and full alternator loads, the horsepower output requirements of the engine shall not exceed the continuous duty rating of the engine. The engine shall be furnished complete with all accessories required for operation, including a fuel tank of sufficient capacity for 8 hours of normal operation, a fuel line sediment bowl, a fuel pump, an air cleaner, a starting battery with an alternator of not less than 65-ampere rated capacity, and an externally adjustable governor capable of limiting the engine speed to the required pump output range. The engine fuel tank shall be mounted outside the engine and pump cabinet with the fuel tank fill pipe located for ease of filling. The engine fuel and air intake systems shall provide adequate fire protection, not less than that provided for a vehicle propulsion system under paragraph 4-3.3 of NFPA No. 407. All electrical leads and circuits shall be properly insulated and shielded to preclude the ignition of engine fuel vapors or product fuel vapors.

* 3.6.14.1.1 Exhaust system. The exhaust system shall conform to paragraph 4-3.4 of NFPA No. 407. A spark arrester conforming to MIL-A-27302 shall be provided on the outside of the cabinet with the exhaust pipe pointing downward and to the rear. The muffler and exhaust pipe shall be covered with a metal shield to guard against contact by operating personnel. The exhaust pipe inside the cabinet shall be wrapped with fire resistant material. Adequate ventilation shall be provided in the engine cabinet. This may include a provision whereby the cabinet doors must remain open when the engine is in operation.

* 3.6.14.2 Pump. An engine driven, self-priming, centrifugal, aircraft fuel dispensing type pump shall be provided. Pump materials shall be corrosion resistant. The pump and pumping system shall provide the performance specified in 3.6.14.3, when handling fuel conforming to MIL-T-5624. The pump shall be capable of producing a vacuum of not less than 18 inches of mercury (Hg) when loading the tank from an outside source or when defueling an aircraft. The pump shall be so constructed and positioned that fuel will not enter into any gearbox or transmission. A drain plug or valve shall be provided for completely draining the pump.

* 3.6.14.3 Pumping system operation. The pumping system shall provide the pressure control specified herein for any servicing rate between zero and the maximum rated capability for the specified

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semitrailer type (see 1.2), based on the aircraft demand for pressure during underwing or closed-circuit refueling. For overwing (open port) refueling, the flow rate shall be controlled by the trigger control of the overwing nozzle. The pumping system shall exhibit no instability in pressure or flow rate under any operating condition. The vacuum of the pump suction at the maximum refueling rate shall be not more than 7 inches Hg, as the tank approaches empty. The 1 1/2-inch hose and reel shall be dedicated to overwing (open port) fuel servicing operations. The 2 1/2-inch hoses and reels shall be dedicated to pressure (underwing or single point) servicing, including closed-circuit refueling of helicopters.

3.6.14.3.1 Type I pumping system operation. The pumping system shall function as follows for type I refuelers:

(a) Refuel the aircraft from the tank through the filter/separator, contamination monitor, meter, 1 1/2-inch servicing hose, and 1 1/2-inch overwing nozzle at any flow rate of up to 100 gpm. The flow rate shall be regulated by control of the overwing nozzle. (See 3.6.22.1 for spout extension required for deep tanks.)

(b) Refuel the aircraft from the tank through the filter/separator, contamination monitor, meter, 2 1/2-inch pressure servicing hose, and underwing nozzle at any flow rate of 0 to 300 gpm, while automatically maintaining 50 $\pm$ 5 psig nozzle pressure at the inlet side of the nozzle. The static pressure at the inlet to the underwing nozzle shall not exceed 60 psig. Once the pumping system controls have been adjusted, the flow shall be regulated on demand ahead of the underwing nozzle. The system shall automatically recover from an externally induced no-flow condition to full flow within 15 seconds after the full flow demand condition has been reestablished. During pressure refueling operation, the pumping system shall limit the total pressure at the inlet to the underwing nozzle to 120 psig at any initial flow rate of 0 to 300 gpm when the flow is shut off in 0.5 second.

(c) Refuel helicopters equipped with a closed-circuit inlet receptacle from the tank through the filter/separator, contamination monitor, meter, 2 1/2-inch hose, and the closed-circuit nozzle adapter assembly at any flow rate of 0 to 100 gpm, while automatically maintaining 15 $\pm$ 2 psig nozzle pressure at the helicopter skin.

(d) Defuel from the aircraft through the overwing nozzle, 1 1/2-inch hose, filter/separator, contamination monitor, and into the tank at flow rates of not less than 60 gpm.

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(e) Defuel from the aircraft through the 2 1/2-inch underwing nozzle, 2 1/2-inch servicing hose, meter, filter/separator, contamination monitor, and into the tank at flow rates of not less than 80 gpm and not more than 100 gpm.

(f) Defuel from the aircraft through the 2 1/2-inch underwing nozzle, 3-inch by 10-foot hose lengths, suction stub, filter/separator, contamination monitor, and into the tank at flow rates of up to 150 gpm.

(g) Self-load through the suction stub at a flow rate of not less than 300 gpm. Self loading shall be accomplished through use of the suction stub specified in 3.6.24 and the 3-inch by 10-foot hoses specified in 3.6.21.5.

(h) Recirculate contaminated fuel through the filter/separator, contamination monitor, meter, hose reel, and into the tank through the 2 1/2-inch hose, underwing nozzle, and bottom loading adapter at flow rates of up to 100 gpm.

(i) Recirculate contaminated fuel through the filter/separator, contamination monitor, meter, hose reel, and into the tank through the 1 1/2-inch hose, overwing nozzle (with converter adapter attached) and bottom loading adapter at flow rates of up to 60 gpm.

* 3.6.14.3.2 Type II pumping system operation. Pumping system operation for type II semitrailers shall be the same as for type I, except that dual 2 1/2-inch hoses may be used simultaneously for pressure refueling. The dual hose configuration shall provide a nominal 600 gpm maximum flow rate capability. In subparagraph (b) of 3.6.14.3.1, where the refuel flow rate for type I semitrailers is specified as 300 gpm, type II semitrailers shall provide a nominal 600 gpm.

* 3.6.14.4 Pump bypass valve and line. An automatic bypass valve and line shall be an integral part of the flow control system. The bypass valve shall serve to divert excess fuel flow to the inlet side of the pump to control fuel pressure. The bypass line shall be located upstream of the filter/separator and contamination monitor. The bypass shall prevent engine speed changes of more than 300 revolutions per minute (rpm) between the full-flow and no-flow for any pumping mode, except defuel.

* 3.6.14.5 System tank return line. In addition to the pump bypass, a tank return line shall be provided. The return line shall allow return of fuel to the tank during defuel and during use of the suction stub. The return line shall be tied in with the bottom loading emergency valve for tank fill shutoff control during defuel operation. The system tank return line shall be located upstream of the filter/separator and the contamination monitor.

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* 3.6.14.6 Check valves. The system tank return line, as well as any other piping or tubing between the tank and pumping system except the tank-to-pump suction line, shall incorporate check valves to prevent backflow from the tank. The check valves shall comply with 49 CFR, section 178.340-8 of MC 306. Check valves or manually controlled valves shall also be located at the inlet and outlet sides of the filter/separator and the contamination monitor to allow routine servicing and replacement of filter/separator elements and fuel quality monitor elements without draining adjacent piping.

* 3.6.14.7 Deadman pressure control valve. A deadman pressure control valve shall be furnished as an integral portion of the flow control system to operate in conjunction with the pump bypass valve and line to control flow rate and pressure (see 3.6.26.2).

* 3.6.14.8 Air compressor. An engine driven, air cooled, air compressor shall be furnished. The air compressor shall be driven by the auxiliary engine which drives the product fuel pump. The compressor shall supply air to the semitrailer air brake reservoir at a sufficient rate to maintain the air pressure required to operate the pneumatic control system.

* 3.6.15 Filter/separator. A filter/separator of aluminum or stainless steel construction with a rated capacity of not less than 300 gpm for type I semitrailer and 600 gpm for type II semitrailer shall be provided and mounted in one of the underslung cabinets in such a manner that water removal and filter/coalescer elements are readily accessible for servicing and replacement. The filter/separator shall be certified by the filter/separator manufacturer to meet the performance requirements of MIL-F-8901. The filter/separator shall incorporate not less than 10 filter/separator coalescer elements conforming to MIL-F-52308 at a rating of not more than 20 gpm each. All aluminum parts shall be anodized in accordance with MIL-A-8625. Inlet and outlet connections shall be not less than 3-inch grooved couplings (Victaulic or equal). The filter/separator unit and components shall be mounted to minimize vibration, oscillation or swing of the various components. The air bleed line to the tank shall be compatible with the product fuel. The filter/separator shall be so installed in the system that all fuel during fueling and defueling operations will pass through the filter/separator. Fuel shall pass through the filter/separator in one direction only, under any condition.

3.6.15.1 Pressure vessel. The filter/separator pressure vessel shall be in accordance with MIL-F-8901 and ASME Boiler and Pressure Vessel Code, section VIII. The pressure vessel shall be designed for a

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working pressure 1 1/2 times the maximum pressure encountered during operation of the pumping system. The pressure vessel shall be hydrostatically tested at 1 1/2 times the working pressure. The pressure vessel flange gasket shall be of the O-ring type. The main vessel shall be provided with lifting devices capable of lifting the entire filter/separator assembly.

* 3.6.15.2 Air eliminator. An air eliminator or constant bleed line shall be provided to allow automatic purging of air from the filter/separator. The air eliminator shall be vented into the bottom of the cargo tank. A check valve shall be provided in the air eliminator line to prohibit the back flow of product fuel. The air eliminator shall be of the manufacturer's recommended size to allow proper air elimination but shall be not less than 3/4 inch in diameter.

* 3.6.15.3 Water slug control. A water slug control shall be provided. The water slug control shall stop the fuel flow when a predetermined water level is reached in the pressure vessel and shall then permit fuel flow when the water level is lowered to a predetermined level. The water slug detector shall be capable of being positively tested by exposing the detector probe to water, without contaminating existing fuel in the tank, and without removing the detector probe from the vehicle. The water slug detector system shall be a Parker Hannifin part number F716 or equal.

* 3.6.15.4 Automatic and manual water sump drain. An automatic water sump drain shall be furnished and shall be either an aluminum or stainless steel vessel or an integral part of the filter/separator. The sump drain shall be provided with an automatic water discharge and a manually operated drain. The automatic water sump drain shall discharge accumulated water from the filter/separator. Provisions shall be made to discharge accumulated water to the ground. The water sump shall be structurally supported to preclude vibration on any fitting.

3.6.15.5 Differential pressure gage. The filter/separator shall be connected to a differential pressure gage mounted on the control panel (see 3.6.26.4). The gage shall indicate the pressure drop across the first and second stages of the filter/separator.

3.6.16 Contamination (fuel quality) monitor. A contamination monitor, meeting the requirements of MIL-M-81380, shall be furnished and installed downstream of the filter/separator. The monitor shall contain a built-in surge suppressor that will provide maximum protection to the

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filter/separator in the event of a pressure surge in the system. One set of monitor elements shall be provided. The monitor shall shut down fuel delivery when excessive contaminant levels are reached in the fuel. The monitor shall be provided with a differential pressure gage. The monitor differential pressure gage is not required if the contamination monitor is incorporated in the filter/separator.

* 3.6.17 Recirculation capability. The tank semitrailer shall have the capability of recirculating contaminated fuel through the filter/separator and contamination monitor. The dispensing hose nozzles shall be connected to the bottom loading adapter to perform fuel recirculation (see 3.6.14.3.1(i) and (j)).

* 3.6.18 Product sampling capability. Provision shall be made for product sampling capability at the nozzle. Product sampling equipment shall be furnished to provide product sampling capabiilty (see 3.6.22.3).

* 3.6.19 Meter. A reversible positive displacement meter shall be provided and housed in the streetside underslung cabinet. The meter capacity shall be not less than 300 gpm for type I semitrailer and 600 gpm for type II semitrailer. The meter shall be capable of momentary overspeeding to 125 percent of maximum rated capacity and shall be calibrated for handling aviation turbine fuel conforming to MIL-T-5624. The meter counter shall have a seven figure totalizer and a four or five figure setback indicating full gallons, with figures not less than 3/4 inch high. The numerals in the setback shall be coated with light reflecting material. All components of the meter in contact with the fuel shall be fabricated from aluminum or stainless steel. The meter shall be located to facilitate reading of numerals, and shall conform to the applicable requirements of the National Bureau of Standards Handbook 44.

* 3.6.20 Piping and couplings. All piping shall be tubing with long sweep bends or pipe with long sweep elbows and shall be of stainless steel or aluminum. Piping shall be isolated from direct contact with dissimilar metals at the point of connection with the tank body or level sensor controlled automatic flow control valve. Tubing, pipe, gaskets, and fittings shall not contaminate the fuel or decompose from contact with fuels. The pumping system piping layout shall be designed to facilitate maintenance on the pumping system and removal of components with minimum disassembly of piping. Piping sections shall be easily removable for maintenance and repair. Where required, piping connections shall be of the grooved-end (Victaulic or equal) type. Piping shall be installed to withstand expansion and vibration without failure and shall conform to 49 CFR, section 178.340-8 of MC 306. Victaulic or equal couplings shall be used only on pipe (not on tubing).

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* 3.6.20.1 Grooved-end pipe clamp couplings or flanges. Grooved-end pipe clamp couplings or flanges, in accordance with MIL-C-10387, shall be provided on the inlet and outlet sides of the pump, meter, valves, filter/separator, contamination monitor, and all other locations where flexible couplings are required.

* 3.6.20.2 Pipe grooves for couplings. Pipe grooves for couplings shall be in accordance with dimensions specified by MIL-P-10388, or the coupling manufacturer. The groove and outside diameters shall be concentric to each other. The groove and the ends of the pipe shall have a chamfer of 0.003 to 0.005 inch. The outside diameter of pipe between groove and end of pipe shall be smooth without evidence of scale, seams, weld splatter, digs or scratches. Fitting gaskets shall be capable of service in minus 20°F to plus 180°F without evidence of deformation or change in dimensions.

* 3.6.20.3 Drains. All sections of the fuel discharge system, including the pump and meter shall be equipped with marked drain plugs located to assure complete drainage of the system.

3.6.21 Hoses and reels.

* 3.6.21.1 Type I dispensing hoses and reels. Two manually driven dispensing hose reels shall be provided. The hose and reels shall be of commercial aviation refueler quality. Gear ratios (ring gear and pinion) shall be designed for easy manual rewinding. One hose reel shall be provided with 75 feet of 1 1/2-inch inside diameter (i.d.) noncollapsible hose, conforming to paragraph 3-2 of NFPA No. 407, suitable for suction defueling at 60 gpm. The second hose reel shall be provided with 50 feet of 2 1/2-inch i.d. noncollapsible hose conforming to paragraph 3-2 of NFPA No. 407, suitable for pressure fuel servicing and suction defueling at 100 gpm. The hose reels shall be mounted in the underslung cabinet on the street side of the semitrailer. The reels shall be of multiwrap or grooved design. The hose reel inlet connections shall be grooved. Provisions shall be made for properly stowed and secured hose nozzles after each hose reel has been rewound and is not in use. Hose reel components coming in contact with fuel shall be nonferrous materials. A nozzle holder with an air-operated brake interlock shall be provided to prevent semitrailer movement when the underwing hose nozzle is not in place, in accordance with paragraph 4-9.10 of NFPA No. 407. A guarded toggle switch shall be provided on the control panel to override the nozzle brake interlock. The switch shall be identified as to function and switch position.

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* 3.6.21.2 Type II dispensing hoses and reels. Hoses and reels for type II semitrailers shall be the same as for type I, except that two 2 1/2-inch noncollapsible hoses and hose reels (capable of simultaneous use for fuel servicing) shall be provided, in addition to the 1 1/2-inch hose and reel.

* 3.6.21.3 Emergency dry break connector. An emergency dry break connector of the frangible or shear pin type shall be provided between the hose end dry break and fuel nozzle dry break couplings of the 2 1/2-inch hose. The dry break shall permit the vehicle to drive away in the event of fire, leaving the sealed off nozzle connected to the aircraft. Both halves of the connector shall seal instantly on breakaway, eliminating fuel spillage from either the nozzle or the delivery hose. The connector shall separate when subjected to an emergency load of not less than 160 pounds and not more than 240 pounds. It shall be possible for the separated connector halves to then be manually removed from the nozzle and hose, and the hose reconnected to the nozzle using the dry break couplings of 3.6.21.4. Normal fueling operations shall then be possible without the emergency breakaway provision. Shear pin replacement shall permit re-use of the emergency dry break connector. The emergency dry break connector shall be similar or equal to Aeroquip Corp. part number AE82129R.

* 3.6.21.4 Dry break coupling. A dry break coupling with a 60-mesh strainer shall be provided for the 2 1/2-inch hoses at the nozzle end. The dry break coupling shall permit full flow during fueling operations and shall seal both the hose half and the nozzle half prior to disconnection, therey preventing the fuel spillage from either the nozzle or the hose. The coupling halves shall be incapable of connection or disconnection until the valves are in the closed position. The coupling valves shall be capable of being easily closed against fuel pressures of 50 psig while the coupling halves are connected and locked. The strainer shall be contained in the nozzle adapter half so as to prevent foreign matter from bypassing the strainer, and to prevent damage or loss during disconnection and inspection. The flange adapter for D-1 type nozzles shall provide an O-ring seal for mating of the adapter to the nozzle. The dry break coupling shall incorporate a positive lock mechanism to prevent disconnectrion through mishandling or through dragging the delivery hoses in either direction during fueling operations. The dry break coupling shall be compatible with the emergency dry break connector of 3.6.21.3. The coupling halves shall be capable of manual connection, positive locking, and disconnection without the use of tools. The coupling halves shall be similar or equal to Aeroquip Corp. part numbers AE82100R and AE82096T (hose half).

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* 3.6.21.5 Noncollapsible hoses and suction stubs. Four 10-foot lengths of 3-inch inside diameter noncollapsible hose, with camlock quick couplers, shall be provided and placed in the hose tubes (see 3.6.10.1). Hoses furnished shall be in accordance with the requirements of 49 CFR, section 178.340-8 of MC 306. Three suction stubs, each 36 inches long with a 90 degree elbow having 3-inch camlock quick couplers shall be furnished. One suction stub shall be fabricated from 2 3/8 inches outside diameter tubing, one from 1 7/8 inches outside diameter tubing and one from 1 inch outside diameter tubing. All tubing shall be nonsparking. The suction stubs shall be used for filling underground tanks, for drafting from drums and underground tanks, and for defueling overwing aircraft (when high defuel rates are required and permitted by the aircraft fuel system).

3.6.22 Nozzles.

* 3.6.22.1 Overwing nozzle. One 1 1/2-inch nozzle conforming to MIL-N-87963, type MD-3, shall be provided for the 1 1/2-inch servicing hose. The overwing nozzle assembly shall permit interchange with the underwing nozzle by utilizing a dry break coupling (Aeroquip Corp. AE82102R, or equal). A screw-in nozzle extension of noncollapsible teflon hose shall be provided with the nozzle to provide defueling capability for helicopter fuel tanks with depths of up to 30 inches.

* 3.6.22.1.1 Convertor adapter. A convertor adapter, similar or equal to Aeroquip Corp. part number AE96673R, shall be provided to convert the 1 1/2-inch overwing nozzle to a 2 1/2-inch D-1 type pressure fueling nozzle. The adapter shall permit underwing emergency refueling (only) and shall permit connection to the bottom loading adapter during fuel recirculation using the 1 1/2-inch hose and reel.

* 3.6.22.2 Closed-circuit nozzle. One closed-circuit nozzle adapter assembly meeting the 15 psig pressure regulating requirements of MIL-N-52747 shall be provided for the 2 1/2-inch servicing hose. The outlet end of the nozzle adapter assembly shall be capable of connection to the fuel inlet receptacle specified in MIL-N-52747 as the helicopter-mounted closed-circuit fuel servicing adapter. The inlet side of the nozzle adapter shall be capable of attachment to any underwing pressure servicing (D-1 type) nozzle, and shall be attached to the underwing nozzle specified in 3.6.22.3 when performing closed-circuit refueling. The nozzle adapter assembly shall be similar or equal to Aeroquip Corp. part number AE82287T.

* 3.6.22.3 Underwing pressure servicing nozzles. An underwing pressure servicing nozzle conforming to either the functional and performance requirements of MIL-N-5877 (designated type D-1), or a

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commercial aviation refueling nozzle capable of connection to the International Standard bayonet adapter, shall be furnished for each 2 1/2-inch servicing hose. Provisions for a product sampling fitting shall be incorporated in the nozzle.

* 3.6.23 Hose reel shutoff valves. A shutoff valve shall be located upstream of each hose reel. The shutoff valves shall be pneumatically controlled by individual controls located on the pumping system control panel. The valves shall open when the control is placed in the ON position and shall remain closed (pressure-tight) at all other times.

* 3.6.24 Suction stub. The tank-to-pump suction line shall incorporate a suction stub with a manually operated shutoff valve. The stub shall be a 3-inch size and shall be provided with a 3-inch camlock quick coupling and dust cap in accordance with MIL-C-27487. The suction stub and shutoff valve shall terminate in the cabinet housing the defuel system, and shall be easily accessible to an operator standing on the ground.

* 3.6.24.1 Suction stub strainer. The suction stub line shall incorporate a nonferrous type 40 mesh strainer. THe strainer shall be rigidly constructed and adequately sized to allow the rated flow at minimum pressure drop. The screen shall be easily removable for cleaning and inspection.

* 3.6.25 Product flow control system. The control system shall automatically provide the pressure control and flow relief functions to accomplish the performance requirements specified herein. The control system shall prevent pump speed surges, pressure surges, and any other type of pumping system instability. The system shall automatically regulate to the required flow conditions, as determined by the overwing nozzle and the aircraft fuel system ahead of the pressure servicing nozzle(s). The control system shall include a pump bypass valve, pressure sensing venturi, dispensing hose reel shutoff valves, filter/separator water discharge slug control, defuel control valves, and operating controls. The pressure sensing venturi shall be located downstream of the filter/separator.

* 3.6.26 Pneumatic control operation. The control system shall be pneumatically operated from the semitrailer air brake system. Safety provisions shall be incorporated that will prevent loss of air in the air brake system in the event of failure of any portion of the control system. Components which operate with fuel and vehicle air under pressure shall be so designed that leakage of fuel into the air system or leakage of air into the fuel system cannot occur. Where possible, components or valves which operate with only one medium (either fluid or

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pneumatic) shall be used. The pneumatic power system shall be essentially a static system which releases air only when controls are operated and only to the extent necessary to purge components. Under no condition shall air be vented into the tank. The bottom loading system pneumatic and fuel components shall be protected with screens, filters, or other devices to prevent contaminants from entering critical components. All pneumatic controls shall be compatible and designed to provided reliable operation in the available air pressure range. The interlock specified in 3.6.11.3 may be pneumatic. All systems and controls shall be independent of the truck tractor. The semitrailer shall be capable of all fuel servicing functions with the truck tractor detached from the semitrailer. All pneumatic lines shall be secured to the tank or frame and protected from damage. All lines exposed to possible ground or brush contact shall be protected with skid plates.

* 3.6.26.1 Operating controls. The operating controls shall include a selector valve, throttle control, deadman control, dispensing hose reel shutoff controls, and operating and emergency controls for the automatic flow control valve.

* 3.6.26.1.1 Selector valve. A selector valve shall be the primary operating control of the pumping system. The selector valve shall have positions for refuel and defuel. The selector valve shall provide for automatic adjustment of the pumping system controls to accomplish the corresponding pumping mode. The control for the automatic flow control valve shall automatically revert to the closed position when the selector valve is placed in the defuel position.

* 3.6.26.2 Deadman control. A deadman control shall be provided and shall be of the air-operated type. The deadman control shall consist of not less than a 50-foot air hose with a deadman control at the free end and shall be stored on a spring loaded reel. The deadman control shall provide control of the pressure control valve (see 3.6.14.7) and shall prevent operation of the pressure control valve unless the deadman control is operated. The deadman control reel shall be located in the streetside underslung cabinet and shall be so designed as to prevent overrunning at end of rewind. Means shall be provided to deactivate the deadman control for overwing refueling, where flow control is provided through the trigger control on the nozzle. The means may be keyed as to which nozzle is out of its holder, such that while the underwing pressure servicing nozzle is lifted, the deadman system is operative.

* 3.6.26.3 Emergency controls. Remote, emergency, quick acting controls for closing (but inoperable for opening) the automatic flow control valve shall be provided in accordance with NFPA No. 407, paragraph 4-8.2. Three emergency controls shall be provided: one near

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the front of the tank on the streetside, one near the rear of the tank on the curbside, and one accessible from the catwalk. The controls shall have fusible devices to insure automatic closing of the valve in case of fire, in accordance with paragraph 4-8.4 of NFPA No. 407. Placarding for each emergency control shall comply with paragraph 4-8.3 of NFPA No. 407.

* 3.6.26.4 Control panel. The pumping system shall incorporate a control panel on the street side of the semitrailer in the pumping compartment. The control panel shall be located within easy reach and shall be clearly visible to an operator standing on the ground. The control panel shall be easily removable or accessible from the rear for maintenance purposes. The control panel shall contain the automatic flow control valve controls, engine controls (including the throttle speed selector control), engine instruments, selector valve control, hose reel shutoff valve controls, underwing or pressure servicing nozzle pressure gage, filter/separator differential pressure gage, pump discharge pressure gage, and pumping compartment light switch. An override toggle switch (to release brake interlocks for emergency driveaway) shall be provided. The controls shall be arranged for maximum convenience to the operator. All controls shall be protected from damage and inadvertent operation.

* 3.6.26.4.1 Engine controls. Engine controls shall include a key-operated ignition and a lockable governor regulated speed selector control (throttle).

* 3.6.26.4.1.1 Throttle. A deadman operated throttle control shall be furnished and shall be governor regulated. The control shall be of reliable design to provide repeated positive and smooth operation on demand. The throttle speed selector control shall be marked to indicate the dispensing rate applicable: "THROTTLE - 100 gpm - 300 gpm - 600 gpm." (The "600 gpm" mark shall only apply to type II semitrailers.)

3.6.26.4.2 Engine instruments. Engine instruments shall include, but shall not be limited to, the following:

(a) Fuel gage
(b) Oil pressure gage
(c) Tachometer
(d) Hourmeter, with a totalizing mechanism of not less than 9,999 hours.
(e) Ammeter, voltmeter or charging indicator
(f) Engine coolant temperature gage.

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* 3.6.26.4.3 Pressure servicing nozzle pressure gage. The nozzle pressure gage shall continuously monitor pressure at the venturi throat. The gage shall read a corrected pressure which simulates the actual pressure at the nozzle. The pressure gage shall have non-ferrous internals, be shock mounted and marked "UNDERWING NOZZLE PRESSURE." The gage shall be of adequate size and located so as to be readily visible during refueling. The gage shall have a compound scale with a range from minus 30 psig to plus 160 psig, and shall be accurate to within one percent of the full-scale range.

* 3.6.27 Grounding cable and reel. A static grounding cable and reel shall be provided. The grounding cable shall consist of not less than No. 12 American Standard Wire Gage stranded wire, clear plastic covered, or 3/32 inch diameter 7x7 stainless or galvanized steel, covered with petroleum resistant rubber or nylon. The cable shall be not less than 75 feet long and shall terminate in a wye. Each leg of the wye shall be 20 feet long. Each end of the wye cable shall be provided with both a heavy duty plier type ground clamp conforming to MIL-C-83413, part number M83413-1, and a plug for plug and jack type bonding to aircraft. The cable shall be contained in a spring loaded, interrupted gear, ratchet type reel, located in the underslung cabinet. A cable outlet guide shall be provided to prevent the grounding cables from tangling or kinking when being extended or retracted in any direction not exceeding 90 degrees from axis of cable outlet guide. A cable stop, which will prevent damage to the clamps due to rapid rewind, shall be provided. The free end of the cable shall be provided with an elastomeric bumper (ball). The ground cable reel mounting surfaces shall be free of paint and contamination, and shall be securely mounted to the cabinet so as to preclude the cable reel assembly from being pulled away from its mountings. A warning plate shall be attached to the cable reel stating: "WALK BACK GROUNDING CABLE TO REEL."

* 3.6.28 Fire extinguisher brackets. In conformance with paragraph 4-3.10 of NFPA No. 407, brackets for two 30-pound dry chemical extinguishers shall be furnished. The extinguishers will conform to O-E-915, size 30, type II, class 2. Each mounting/carrying bracket shall be easily and quickly operated. One fire extinguisher bracket shall be mounted vertically on each side of the semitrailer. Protection shall be provided from splashing of mud and water. The fire extinguishers, when installed by the Government on the brackets furnished, shall be readily accessible to personnel standing on the ground.

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3.6.29 Operational markings.

3.6.29.1 Instruction plates. Metal instruction plates shall be permanently attached and located as to be accessible and visible to the operator.

* 3.6.29.2 Diagrams. Schematic diagrams of the piping, including the bottom loading system, pneumatic system, and electrical system, shall be provided. Each valve, switch, and other pertinent component on the diagram shall be properly identified to correspond to the markings on like parts of the semitrailer. Schematic diagrams shall be provided on metal plates or plastic encapsulated plates and shall be of a material which will be durable and remain legible for the life of the equipment. Valves that are normally closed or normally open shall be identified as to their normal mode. Diagrams shall be located and attached as specified in 3.6.29.1.

* 3.6.29.3 Operating instructions. Operating instructions shall be clearly and concisely stated. All designations used in the operating instructions shall be the same as shown on the schematic diagrams (see 3.6.29.2). Operations which affect safety of the pumping system shall be listed under the title "CAUTION" at the top of the plate. Operations which affect safety of the operator shall be listed under the title "WARNING" at the top of the plate. The words "CAUTION" and "WARNING" shall be larger than the operational instructions. Specific instructions shall be listed separately for each pumping system operating mode, with the pumping system mode as the title for the various sections on the plate. At least the following notice shall be included: "WARNING: AUTOMATIC FLOW CONTROL VALVE CONTROL MUST BE IN CLOSED POSITION WHEN BOTTOM LOADING."

* 3.6.29.4 Bottom loading plate. Separate, concise, step-by-step operating instruction plates for the bottom loading system shall be permanently mounted on or adjacent to their respective controls.

3.6.29.5 Identification plates. Valves, switches, levers, and controls used in the operation of the semitrailer shall be properly identified by nonferrous identification plates. The data plates shall incorporate a numerical or alphabetical system of identification keyed to the schematic and an indication of the relative position of the lever or control handle, such as "ON" or "OFF."

3.6.29.6 Filter/separator element change criteria data plate. An element change criteria data plate shall be mounted adjacent to the filter/separator pressure gage specified in 3.6.26.4.3. The data plate shall be entitled "FILTER/SEPARATOR ELEMENT CHANGE CRITERIA." The plate shall indicate that the elements shall be changed at 20 psi pressure

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differential, every 2 years, or when the contamination monitor triggers (shuts off the flow), whichever occurs first.

3.6.29.7 Contamination monitor element change criteria data plate. A data plate, similar to the plate described in 3.6.29.6 shall be provided. The data plate shall be entitled "CONTAMINATION MONITOR ELEMENT CHANGE CRITERIA" and shall describe the criteria for changing of the contamination monitor elements. The plate shall be located in close proximity to the filter/separator pressure gage specified in 3.6.26.4.3.

* 3.6.29.8 Fuel load chart. A temperature-compensated chart for converting inches of depth in the tank to gallons of fuel in the tank shall be provided. The chart shall be permanently installed in the streetside underslung cabinet so as to be readily visible. The chart shall be fabricated of a material so as to remain legible for the life of the vehicle.

3.6.30 Lubrication. Lubrication means shall be provided for all parts of the equipment normally requiring lubrication. Where the use of high lubricating pressure will damage grease seals or other parts, fittings with pressure release shall be used.

3.6.31 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government inspector, the contractor shall service and adjust the semitrailer for immediate operational use. The servicing and adjusting shall include at least the following: inflation of all tires, adjustment of brakes, proper functioning of all lighting, and complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point. The semitrailer shall be conspicuously tagged to identify the lubricants and their temperature range.

3.6.32 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

3.6.33 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welding shall not be employed as a repair measure for defective parts. Weld design and application shall be such as to provide adequate transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Sequence of tests. The road test shall be the first test performed. The sequence of tests given herein is recommended; however, the manufacturer may modify the sequence in which the remaining tests are performed, subject to approval by the Government.

* 4.2 Cleaning. Prior to the beginning of any tests, all vehicles, including the first production vehicle, shall be subjected to the following cleaning procedure: Each vehicle shall be carefully inspected for evidence of contaminants in the tank and fuel system, such as loose, spattered or excess solder, metal chips, or other debris. Test fluid, as specified in 4.6.2, shall be pumped through the system as specified in 3.6.17, for a minimum of 20 minutes at not less than 100 gpm with all filter elements in place. At the end of the 10 minutes, strainers and low point drains shall be examined for evidence of contaminants. If any trace of contaminants is found, the recirculation of test fluid at 100 gpm shall continue for 10 minute periods until all trace of contaminants has been removed. Strainers and low point drains shall be examined at the end of each 10 minute period. When all trace of contaminants has been removed, new elements shall be installed in the filter/separator and contamination monitor, and utilized in all performance tests specified herein. After flow tests are completed, a new set of elements shall be installed.

* 4.3 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract. The first production vehicle shall be examined and tested as specified in table II. The sequence of test events may be changed if required to reduce testing time, provided that the road tests are run before functional tests, and the contracting officer approves the altered sequence.

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* 4.4 Production vehicle inspection. Each production vehicle shall be examined and tested as specified in table II.

TABLE II. Inspection requirements for first production and production vehicles.

Paragraph	Description	First production	Production
4.5	Examinations	X	X
4.5.1	Vehicle weight	X	
4.5.2	Radio interference suppression examination	X	X
4.6.4	Road test	X	
4.6.4.1	Service brake test	X	
4.6.4.2	Parking brake test	X	
4.6.4.2.1	Parking brake interlock test	X	X
4.6.5.1	Tank air test	X	X
4.6.5.2	Vent operation test	X	X
4.6.5.3	Piping system hydrostatic test	X	X
4.6.5.4	Meter flow rates	X	X
4.6.5.5	Hose reel strength test	X	
4.6.6.1	Bottom loading system test	X	X
4.6.6.2	Dispensing flow test	X	X
4.6.6.3	Automatic valve closure	X	
4.6.6.4	Defuel test	X	X
4.6.6.5	Pump suction and self-load test	X	
4.6.7	Test for radio interference suppression (see 4.5.2)	X	
4.6.8	Certification	X	X

4.5 Examinations. Each semitrailer shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with the specified requirements or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5.1 Vehicle weight. The weight on the fifth wheel and axles shall be measured with the semitrailer standing on a flat surface and empty. The weight on the fifth wheel and axles shall also be measured with both the cargo tank and the pumping system fully flooded (as during normal operation) with test fluid. The imposed loads on the fifth wheel

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and axles shall be used to ascertain that the suspension, axles, and tires furnished are of adequate capacity to meet the requirements of 3.2.2 and 3.2.4.

4.5.2 Radio interference suppression. In lieu of the test specified in 4.6.7 and with the approval of the contracting officer, a certification of conformance to SAE J551 shall be submitted by the manufacturer. The vehicle shall be examined to verify the installation of the radio interference suppression devices specified in the certification.

4.6 Tests.

4.6.1 Test apparatus. Apparatus used in conjunction with testing specified herein shall be of laboratory precision type and shall have a calibration schedule acceptable to the Government. Any transducers used shall be individually calibrated immediately before tests are run. A calibration record shall be made available to the Government for examination showing that the transducers have been calibrated.

* 4.6.2 Test fluid. The test fluid used for all hydrostatic tests and for all liquid flow tests shall be clean commercial solvent with a viscosity equal to commercial turbine fuel (Jet A), but having a higher flash point and lower vapor pressure.

* 4.6.3 Test data. During testing of the pumping system, at least the following data shall be recorded:

(a) Vacuum at pump inlet - inches Hg.
(b) Pump discharge pressure - psig.
(c) Pressure loss through nozzle - psi.
(d) Fuel flow rates - gpm.
(e) Engine speed - rpm.
(f) Pump speed - rpm.

* 4.6.4 Road test. The road tests shall consist of driving the semitrailer-tractor combination, loaded with rated payload, a distance of not less than 40 miles under the conditions specified in 3.3.2. At least 20 percent of the distance shall be over hard uneven terrain. At least five sudden stops shall be made from a speed of not less than 20 mph to provide dynamic load on tank attachment points. Tracking ability at low and high speeds shall be observed to verify conformance to 3.3.2.2, and turning ability shall be observed to determine conformance to 3.3.2.1. After the test, the semitrailer shall be examined for evidence of misalinement, binding, or other malfunction.

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4.6.4.1 Service brake test. The semitrailer service brakes shall be tested to verify conformance to 3.3.2.3. Breakaway situation shall be simulated to verify braking ability upon breakaway.

4.6.4.2 Parking brake test. The semitrailer parking brakes shall be tested to verify conformance to 3.4.5.2. The 10 percent slope parking brake test shall be conducted with the semitrailer disconnected from the towing vehicle and resting on the landing legs.

* 4.6.4.2.1 Parking brake interlock test procedure. Loading and unloading tests (see 4.6.6.1) shall be performed with the semitrailer attached to the towing vehicle and the parking brakes not manually applied to determine that they are automatically applied in conformance to 3.4.5.2. The adapter/brake interlock for both legs of the bottom loading adapter shall be checked according to the following procedure: the parking brake override control on the control panel shall be set in the OFF position. A 4-inch API coupler shall be connected to the bottom loading adapter. The brakes shall be observed to insure that they apply. The brakes shall then be released and the test repeated with a 2 1/2-inch D-1 nozzle connected to the bottom loading adapter. To check the nozzle-brake interlock, the brakes shall be released and the underwing nozzle removed from its stowage bracket. The brakes shall again be observed to insure that they apply. The override toggle switch on the control panel shall be checked to insure that it overrides the brake interlock of only the underwing nozzle stowage bracket.

4.6.5 Component tests.

* 4.6.5.1 Tank air test. To determine conformance to 3.6.3, the semitrailer cargo tank shall be subjected to a static air pressure test at 5 psig for a period of not less than 5 minutes. A soap solution shall be applied on the exterior of the tank at all welds and joints as an aid in the visual detection of leaks. Vents and other normal pressure relief systems shall be sealed closed during the test.

* 4.6.5.2 Vent operation test. The venting equipment shall be tested for conformance to the requirements of 3.6.8. The vents shall relieve positive and negative pressures and prevent steady stream leakage.

* 4.6.5.2.1 Vent operation and pressure/vacuum holding test. When conformance to 3.6.8.5 is required (see 6.2 and 3.6.8.5), the contractor may use pressure transducers or other industry accepted methods to measure normal venting pressure limits for conformance to 3.6.8.5. A suggested test method suitable for the low pressures involved is as follows:

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(a) Position the first production semitrailer such that a test fluid hose can be connected between the bottom loading adapter and a pressurized source for filling the tank and a hose can be connected to a receptacle for discharging test fluid from the tanks. Partially fill the tank (required for steps (f), (g) and (h) below).
(b) Attach the vapor recovery flange to a U-tube water manometer (mercury unacceptable). (The U-tube manometer can be fabricated from clear plastic tubing mounted on a ruled board.)
(c) Bottom load test fluid into the tank until a positive air pressure differential of 18 inches of water is reached. (Opening the bottom loading valve will open the automatic vents specified in 3.6.8.5 thus pressurizing the vapor recovery line and the manometer.) Leave the bottom loading valve open. Close all other valves. Insure that test fluid is not draining back through the bottom loading check valve.
(d) The pressure decay over a 5 minute period shall be observed for conformance to the 1 inch of water requirement of 3.6.8.5(a).
(e) Bottom load additional test fluid into the tank observing the manometer and the vents to insure that the vents do not open before 18 inches of water is reached and that the vents do open before 27.7 inches of water is reached (see 3.6.8.5(b)).
(f) Drain test fluid from the tank until a negative pressure differential of 6 inches of water is reached and hold at that level.
(g) The pressure increase over a 5 minute period shall be observed for conformance to the 1 inch of water requirements of 3.6.8.5(a).
(h) Drain additional test fluid from the tank observing the manometer and the vents to insure that the vents open before a vacuum of 10.4 inches is reached (see 3.6.8.5(c)).
(i) Similar tests conducted by pressurizing the tank through an air hose connected to the bottom loading flange are not acceptable since vacuum tests on the entire tank cannot be performed using this method.

* 4.6.5.2.2 Emergency venting. The exposed surface area of each tank compartment shall be measured and the capacity of each vent determined by its required stamping and certification (see MC 306). The total venting capacity shall be calculated and compared to the surface area to determine conformance to the requirements of 49 CFR, section 178.341, of MC 306.

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* 4.6.5.3 Piping system hydrostatic test. All components of the piping system, including the automatic flow control valve, all pipe fittings, hoses and hose reels, hose nozzles, and other components through which fuel passes, shall be hydrostatically tested with the test fluid specified in 4.6.2 at 150 percent of design pressure for a period of not less than 30 minutes, in accordance with paragraph 4-10.2 of NFPA No. 407. The system shall be examined for leakage and the hose shall be examined for porosity.

* 4.6.5.4 Meter flow rates. The meter shall be tested at 125 percent of maximum rated capacity, using the test fluid specified in 4.6.2, for a period of not less than one minute. Meter flow rates shall be measured to determine conformance to 3.6.19. Meter accuracy and calibration shall be verified during the bottom loading and dispensing flow tests of 4.6.6.1 and 4.6.6.2.

* 4.6.5.5 Hose reel strength test. Hoses shall be completely unwound and laid out at an angle of 45 degrees from the axis of the semitrailer. The hose shall be rewound a minimum of 20 times. The 45 degree angle shall be maintained during rewind as much as possible in order to bring the hose to bear on the reel disks. There shall be no indication of permanent deformation of the reel disks. The emergency dry break connectors, specified in 3.6.21.3, shall be tested to determine that their tensile strengths meet specified requirements. The connectors or shear pins shall be replaced after the test.

* 4.6.6 System tests. System tests shall be conducted with the semitrailer disconnected from the towing vehicle and resting on the landing legs (to verify independent semitrailer operation in conformance to 3.3.1).

* 4.6.6.1 Bottom loading system test. The bottom loading system shall be tested to determine conformance to the requirements of 3.6.11. Using either leg of the bottom loading adapter (at the manufacturer's option), the tank shall be loaded from empty at a rate of not less than 600 gpm through a calibrated meter. Failure of the tank to accept 600 gpm, failure of the tank to accept less than its required capacity before closing of the bottom loading valve, acceptance by the tank of more than its required capacity (in excess of 1 percent) or liquid flow from the vents shall constitute failure of the test. The capacity marker shall be checked against the meter reading for proper setting. Upon any failure, corrections shall be made and the entire test repeated to establish that all deficiencies have been corrected.

* 4.6.6.2 Dispensing flow tests. To determine conformance to 3.6.14.3 through 3.6.14.3.2, test fluid shall be pumped from the tank to

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a point which is 20 feet above the pump base. The flow rates and nozzle pressures shall be as specified in 4.6.6.2.1 and 4.6.6.2.2. Each flow rate shall be for a minimum period of 8 minutes and shall be through the nozzle and hose specified in 4.6.6.2.1 through 4.6.6.2.2. In all flow tests, a suitable shutoff valve may be installed at the outlet of the nozzle to simulate the 20-foot head. During the flow tests, data shall be recorded in accordance with the test data required in 4.6.3. Unless otherwise indicated, tests for both semitrailer types shall be identical. During the tests, the tank shall be completely discharged through the onboard meter fuel servicing system. The recorded flow out through the onboard meter shall be compared to the amount of fluid pumped into each tank compartment to insure proper calibration of the onboard meter.

4.6.6.2.1 Overwing refuel flow tests. The overwing refuel flow test shall be conducted in accordance with the conditions of subparagraph (a) of 3.6.14.3.1.

* 4.6.6.2.2 Pressure (underwing) refuel flow tests. Pressure (underwing) refuel flow tests shall be conducted in accordance with the conditions of subparagraph (b) of 3.6.14.3.1. The maximum flow rate shall be not less than 300 gpm for type I semitrailer and be not less than 600 gpm for type II semitrailer. Type II semitrailer shall use both 2 1/2-inch hoses in simultaneous operation whenever the 600 gpm flow rate is required.

* 4.6.6.2.3 Closed-circuit (pressure) refuel flow tests. A closed-circuit refuel flow test, using the 2 1/2-inch hose and closed-circuit nozzle adapter assembly of 3.6.22.2 mounted on the nozzle of 3.6.22.3, shall be conducted to verify conformance to subparagraph (c) of 3.6.14.3.1.

* 4.6.6.3 Automatic valve closure. While pumping at the maximum flow rate for the semitrailer type through the pressure servicing nozzle(s) and with the nozzle(s) raised to a minimum of 10 feet above ground level, not less than 10 valve closures shall be made to determine conformance to 3.6.14.3.1(b). The total pressure buildup at the nozzle, including the 50 psig nominal pressure, the 10-foot head, and any surge pressure, shall not exceed 120 psig.

* 4.6.6.4 Underwing defuel test. To determine conformance to 3.6.14.3.1(e), the 2 1/2-inch hose shall be attached to a storage tank using the underwing pressure servicing nozzle. The selector shall be set to the defuel position. Check the position of the automatic flow control valve control for conformance to 3.6.26.1.1. Test fluid as specified in 4.6.2 shall flow from the hose reel into the semitrailer

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cargo tank. A flow rate of not less than 80 gpm and not more than 100 gpm shall be maintained until the tank is filled to capacity. The level of test fluid in the storage tank shall be no higher than 6 feet above the semitrailer pump at the beginning of the test. During defuel tests, the tank fill cutoff system shall be tested to verify that the tank does not overfill during defuel. The test shall be partially repeated with the 1 1/2-inch hose and overwing nozzle to determine conformance to 3.6.14.3(d).

4.6.6.5 Pump suction end self-load test. To determine conformance to 3.6.12, 3.6.14.2 and 3.6.14.3.1(g), the four 10-foot lengths of hose (see 3.6.21.5) shall be connected from the suction stub to a storage tank or other typical fuel source below ground level, such that the total dynamic suction head at 300 gpm, measured at the pump, shall be not less than 18 inches Hg. A flow rate of not less than 300 gpm, and a suction head of 18 inches Hg, shall be maintained for a minimum of 10 minutes.

4.6.7 Test for radio interference suppression. Radio interference characteristics of the vehicle shall be measured in accordance with SAE J551 to verify conformance to 3.1.1.3. The certification and examination specified in 4.5.2 may be furnished in lieu of this test, upon approval of the contracting officer.

* 4.6.8 Certification. A manufacturer's certificate of compliance to MC306 as defined in 49 CFR, sections 178.340 and 178.341 shall be furnished, endorsed by the vehicle contractor if different, and securely stowed and shipped with each semitrailer together with instructions for the recipient regarding the certificate's retention and use. The certificate shall bear, in addition to the signatures specified above, the typed and signed name of the government official accepting the vehicle.

4.6.9 Test records. Contractor's records of all first production and production vehicle tests and inspections, giving the results of said tests and inspections to determine compliance with the requirements of this specification, shall be kept complete and available to the government's representative and the contracting officer. Test and inspection records shall be signed and approved by a person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory satisfactory to the Government capable of conducting required tests to determine compliance with all the requirements of this specification.

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4.6.10 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.6.11 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

* 6.1 Intended use. This semitrailer is intended for use as an on-station fuel servicing unit for all military aircraft, including all helicopters. Fuel transfer modes include pressure fuel servicing, closed-circuit type refueling, and overwing type refueling and defueling. The semitrailer may be used for fuel tansfer either with or without the towing tractor.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of semitrailer required (see 1.2).
(c) Identification of appropriate service for painting and marking (see 3.1.1.1).
(d) Concealed marking, if required (see 3.1.1.1).
(e) Rustproofing, if required (see 3.1.1.2).
(f) Tropical rustproofing, if required (see 3.1.1.2).
(g) Disc wheels, if required (see 3.4.4).
(h) Bias ply tires, if required (see 3.4.4.1).
(i) If a carrier for a spare assembly is not required (see 3.4.4.3).
(j) Spare wheel or rim, if required (see 3.4.4.4).
(k) Spare tire, if required (see 3.4.4.5).
(1) Independent 24-volt lighting system, if required (see 3.4.7.3).
(m) If night servicing lights are not required (see 3.4.7.6).
(n) Simultaneous operation of landing legs, if required (see 3.4.8).

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(o) A 2-inch NPT coupling and plug, if required (see 3.6.7).
(p) Normal venting pressure limits, if required for aviation gasoline tankers (see 3.6.8.5).
(q) Type of bottom loading adapter required, if not "Y" type (see 3.6.11.3).
(r) Three-inch vapor recovery system, if required in lieu of 4-inch (see 3.6.13).
(s) Diesel pump engine, if required (see 3.6.14.1).

6.3 Definitions.

* 6.3.1 Aircraft. An aircraft is defined as any helicopter or any fixed-wing airplane.

6.3.2 Improved road. An improved road is a smooth, hard surfaced road such as a concrete- or asphalt-paved hightay.

6.3.3 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating service having occasional chuckholes and exposed rocks.

6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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